

CLINICAL PROBLEM-SOLVING

Caren G. Solomon, M.D., M.P.H., *Editor*

Confused about Confusion

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In this Journal feature, information about a real patient is presented in stages (boldface type) to an expert clinician, who responds to the information by sharing relevant background and reasoning with the reader (regular type). The authors' commentary follows.

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During morning rounds, the hematologist on call noticed confusion in a 49-year-old female patient. She had been admitted 5 days earlier for CD19-directed chimeric antigen receptor T-cell (CAR-T) therapy with infusion of axicabtagene ciloleucel for relapsed diffuse large B-cell lymphoma. Although the patient had been able to have a normal conversation the previous day, on this morning the hematologist found her to have bradyphrenia and to be disoriented to time and place. She had difficulty writing and counting backward. No signs of headache, vomiting, focal deficits, vision abnormalities, hallucinations, pain, changes in bowel or bladder function, cough, or shortness of breath were noticed. No trauma or seizure was observed. Vital signs, including temperature, were normal. A neurologic examination was notable only for disorientation and for slow speech with difficulty in putting words together, with no focal abnormalities. The rest of her physical examination was normal.

Neurologic abnormalities consistent with an acute encephalopathy developed after the patient received autologous T lymphocytes genetically modified to express chimeric antigen receptors that combine the intracellular signaling part of a T-cell receptor with the antigen-recognition domain of a CD19-specific antibody. The CD19 antigen is expressed on normal and malignant B lymphocytes. Encephalopathy has a broad differential diagnosis. Infection is of particular concern in patients who receive CAR-T therapy because they are profoundly immunocompromised owing to lymphopenia, neutropenia, and hypogammaglobulinemia that result from CAR-T therapy and previous immunochemotherapy. However, the current absence of fever and localizing symptoms in this patient makes infection less likely. Metabolic causes should be ruled out; these include electrolyte abnormalities (e.g., hypercalcemia related to lymphoma), renal or hepatic dysfunction, thiamine deficiency, and substance abuse or withdrawal. I am most concerned about the possibility of immune effector cell–associated neurotoxicity syndrome, which affects two of three patients after treatment with axicabtagene ciloleucel, on average 5 days after infusion, with an initial presentation characterized by acute encephalopathy with disorientation and difficulty in writing and finding words.

Two years earlier, the patient had received a diagnosis of diffuse large B-cell lymphoma. Despite intensive immunochemotherapy, the lymphoma relapsed twice. She had no history of central nervous system (CNS) involvement. She was referred to our

center to be evaluated for eligibility for CAR-T therapy. She had no coexisting conditions. She had previously worked as a social worker, had stopped smoking 2 years before the current presentation, and did not use alcohol or illicit drugs. As part of the evaluation for CAR-T treatment, magnetic resonance imaging (MRI) of the head had been performed 6 weeks before admission; the findings were normal. Standard serologic screening for the human immunodeficiency virus (HIV), *treponema pallidum*, hepatitis B and C, and toxoplasma was negative. During the 4-week CAR-T manufacturing time after the patient underwent leukapheresis, she received methylprednisolone at a dose of 2 g twice daily for 3 days to prevent rapid progression of lymphoma. Apixaban, at a dose of 5 mg twice daily, was started for a catheter-related deep-vein thrombosis. Other medications included pantoprazole (40 mg once daily), valacyclovir (500 mg twice daily for prophylaxis against herpes simplex virus and varicella-zoster virus), and trimethoprim-sulfamethoxazole (80 to 400 mg once daily for prophylaxis against *Pneumocystis jiroveci* pneumonia). She was admitted to the hospital to receive lymphodepleting chemotherapy with fludarabine and cyclophosphamide followed by CAR-T infusion; 24 hours after she received CAR-T infusion, a fever developed without other signs or symptoms. Laboratory studies revealed a C-reactive protein level of 6 mg per liter (normal value, <5), and results of a complete blood count were normal except for a reduced neutrophil count of 1150 per microliter (normal range, 1500 to 7200). The lactate dehydrogenase (LDH) level, coagulation tests, and levels of electrolytes, glucose, creatinine, aminotransferases, and bilirubin were normal. She was treated with acetaminophen, tocilizumab (an interleukin-6-receptor-blocking antibody, at a dose of 8 mg per kilogram of body weight) for presumed cytokine release syndrome, and empirical broad-spectrum antibiotics until blood and urine cultures remained negative. The fever resolved after 48 hours, and the patient's condition was stable until acute encephalopathy developed.

Cytokine release syndrome, a multisystem inflammatory response provoked by in vivo CAR-T proliferation and activation of inflammatory cells, can explain the episode of fever, since cyto-

kine release syndrome occurs in the vast majority of patients after infusion of axicabtagene ciloleucel and fever is a cardinal sign of this syndrome. Cytokine release syndrome frequently develops within the first days after infusion and responds well to tocilizumab. Differentiating infections from cytokine release syndrome is a challenge. During the first month after infusion of axicabtagene ciloleucel, infections occur in approximately one third of patients and are caused mainly by common bacterial pathogens and respiratory viruses. The development of encephalopathy in this patient soon after suspected cytokine release syndrome further supports concern for immune effector cell-associated neurotoxicity syndrome, which is almost invariably preceded by cytokine release syndrome. Because immune effector cell-associated neurotoxicity syndrome can progress quickly, treatment with dexamethasone is advised as soon as this syndrome is suspected and should be started in this patient. If rapid resolution does not occur, a cranial MRI scan should be obtained.

Immune effector cell-associated neurotoxicity syndrome was considered to be the likely cause of the patient's symptoms, and dexamethasone was administered at a dose of 10 mg. Subsequently, her bradyphrenia resolved and she was able to write and count effortlessly. However, 12 hours later, the same symptoms recurred and her level of consciousness declined. She could still follow simple commands but opened her eyes only in response to voice. A cranial MRI showed no abnormalities (Fig. 1). Given the concern for progressive immune effector cell-associated neurotoxicity syndrome, the dexamethasone dose was increased to 10 mg every 6 hours and was continued at that dose for 2.5 days. Seizure prophylaxis was started with levetiracetam (500 mg twice daily). Again, her mental state normalized rapidly.

Both the rapid resolution of symptoms after initiation of treatment with dexamethasone and the early recurrence of symptoms are characteristic of immune effector cell-associated neurotoxicity syndrome. Glucocorticoids are the cornerstone of treatment for this syndrome, with higher doses used in more severe illness. The cranial MRI showed no evidence of cerebral edema or

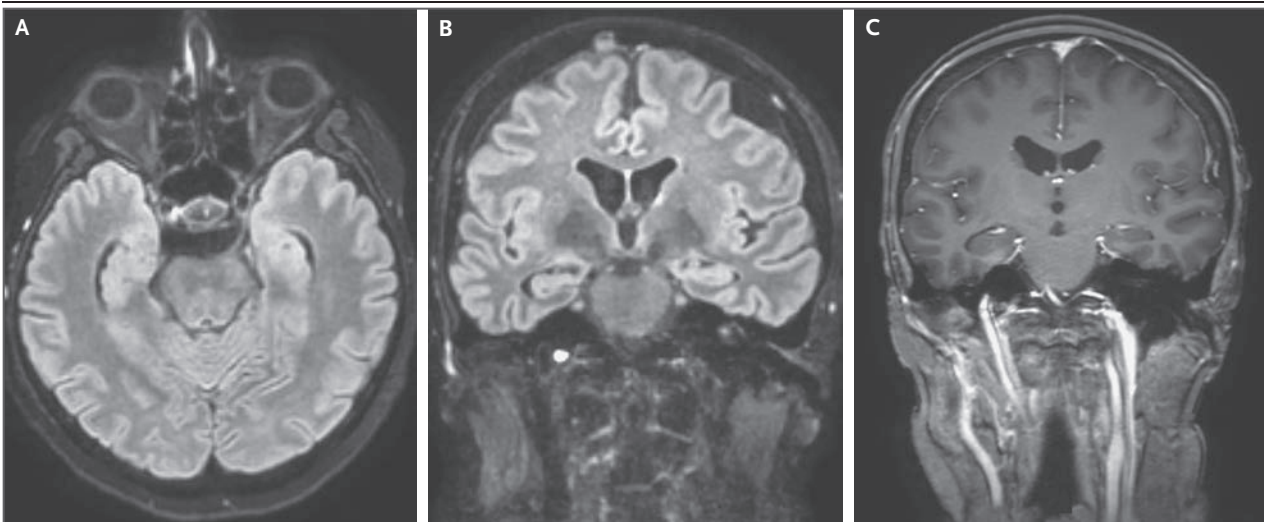


Figure 1. Cranial MRI 7 Days after Chimeric Antigen Receptor T-cell (CAR-T) Infusion.

Axial (Panel A) and coronal (Panel B) fluid-attenuated inversion recovery (FLAIR) images and a coronal gadolinium-enhanced T1-weighted image (Panel C) of the temporal lobes and hippocampal area show no abnormalities.

space-occupying or vascular lesions. If the symptoms had persisted, analysis of cerebrospinal fluid (CSF) would have been indicated to rule out leptomeningeal lymphoma, autoimmune encephalitis, and infectious meningitis or encephalitis, but the rapid improvement in her condition is reassuring. Although seizures may occur infrequently in the context of immune effector cell-associated neurotoxicity syndrome, no seizures were observed in this patient, and evidence is lacking to support the use of prophylactic levetiracetam. Since her condition has normalized and no pathologic features were seen on MRI, dexamethasone should be tapered within days and her neurologic status should be closely monitored, since immune effector cell-associated neurotoxicity syndrome can recur.

The patient remained without symptoms during dexamethasone tapering (at a rate of 10 mg every 2 days, beginning 3 days after dexamethasone was started). Seven days after dexamethasone was started, while the patient was still receiving 10 mg twice a day, recurrent confusion was noted by her nurse. She was restless and reported hearing voices talking about her. She was alert, oriented, and able to maintain attention during conversation, but she was agitated. Her vital signs were normal, she was afebrile, and she had no localizing signs or symptoms.

She is now restless and agitated and has auditory hallucinations and delusions, which suggest an acute psychosis. The new onset of symptoms in the hospital suggests a medical or drug-related cause. Relapse of immune effector cell-associated neurotoxicity syndrome 7 days after she initially presented with the syndrome, while the dose of dexamethasone is still slowly being tapered and with the patient having had no neurologic symptoms for days, is possible but unlikely, especially since psychosis is an uncommon manifestation. Glucocorticoid use can induce psychosis, most commonly in the first week after treatment starts. Levetiracetam use has also uncommonly been associated with neuropsychiatric symptoms. Before pursuing broad-based testing for alternative causes of psychosis, I would discontinue dexamethasone, assess whether the psychosis resolves, and consult a psychiatrist.

A psychiatric assessment revealed a history of depression associated with prolonged bereavement after the death of her husband 15 years earlier but no subsequent psychiatric disorders. Dexamethasone was rapidly tapered and haloperidol (at a dose of 2 mg twice daily) was started for suspected glucocorticoid-induced psychosis. Soon her behavior returned to normal, and 27 days after CAR-T infusion, she was discharged home in good health. One week after discharge, she was

doing well, and ^{18}F -fluorodeoxyglucose positron-emission tomography and computed tomography (CT) showed complete metabolic remission of the lymphoma.

The clinical course in this patient supports the initial diagnosis of cytokine release syndrome followed by immune effector cell–associated neurotoxicity syndrome, which is most often completely reversible, and a subsequent glucocorticoid-induced psychosis.

Two weeks after discharge, however, the patient presented to the emergency department with a recurrence of confusion. She was disoriented to time, place, and person and could not remember her recent admission or having ever been treated for lymphoma. She was able to recognize her daughter. She did not report feeling unwell, and this status was confirmed by her daughter. On physical examination, her heart rate was 116 beats per minute, and her blood pressure, temperature, and respiratory rate were normal. A neurologic exam showed no focal deficits, and the rest of her physical examination was normal.

This new episode of acute encephalopathy 6 weeks after CAR-T infusion is characterized by memory loss. Memory loss can point to limbic encephalitis, which can be caused by viral CNS infections and autoimmune or paraneoplastic inflammation. Although inflammation of limbic structures shown on MRI has been reported in association with immune effector cell–associated neurotoxicity syndrome, the time that has elapsed since CAR-T infusion makes this syndrome unlikely. Given her immunocompromised state, I am now more concerned about the possibility of infection, especially viral encephalitis (e.g., caused by herpes simplex virus, varicella–zoster virus, cytomegalovirus [CMV], Epstein–Barr virus, JC virus, or human herpesvirus 6 [HHV-6]). Fungal infections (e.g., aspergillus species or *Cryptococcus neoformans*) and tuberculous or bacterial meningitis (e.g., *Listeria monocytogenes*) can be considered, but they usually manifest with other symptoms, such as fever, headache, and focal deficits, which are absent in this patient. Toxoplasmosis, HIV, and syphilis are unlikely since serologic tests were negative during CAR-T screening. The patient's travel history and expo-

sure to animals and insects should be reviewed. Cranial imaging is warranted to assess for evidence of CNS infection or inflammation (e.g., hyperintense signals in temporal lobes) or recurrent lymphoma. Analysis of CSF should be performed and should include cultures, flow cytometry, and autoimmune and paraneoplastic antibody testing.

The patient's daughter reported that her mother, who was born in Turkey, had moved to the Netherlands as a child. Her travel history was unremarkable. She had no contact with animals and there was no known tick exposure. Laboratory assessment revealed a blood leukocyte count of 1700 per microliter (normal range, 3500 to 10,000), an absolute neutrophil count of 1000 per microliter, a C-reactive protein level of 0.7 mg per liter, and a ferritin level of 4639 μg per liter (normal range, 15 to 150). Blood levels of hemoglobin, haptoglobin, electrolytes, creatinine, aminotransferases, bilirubin, LDH, and thyrotropin were normal, as were the platelet count and the prothrombin time. A CT scan of the chest and abdomen showed no abnormalities. Blood cultures were obtained and quantitative serum polymerase-chain-reaction (PCR) testing was performed for the presence of CMV, Epstein–Barr virus, HHV-6, herpes simplex virus, JC virus, and varicella–zoster virus DNA.

The pancytopenia and hyperferritinemia are probably the results of an acute-phase reaction and not hemophagocytic lymphohistiocytosis, since the criteria for the latter after CAR-T therapy require ferritin levels higher than 10,000 μg per liter and at least two of the following findings: markedly elevated aminotransferase or bilirubin levels, oliguria or markedly elevated creatinine levels, pulmonary edema, or histologic evidence of hemophagocytosis. The low level of CRP may be explained by recent tocilizumab use and does not rule out an acute-phase reaction.

A cranial MRI showed bilateral swelling of the hippocampus and subtle gadolinium enhancement of the limbic system (Fig. 2). Treatment with foscarnet (at a dose of 90 mg per kilogram twice daily) and ganciclovir (5 mg per kilogram twice daily) was initiated for suspected viral encephalitis. A lumbar puncture, performed

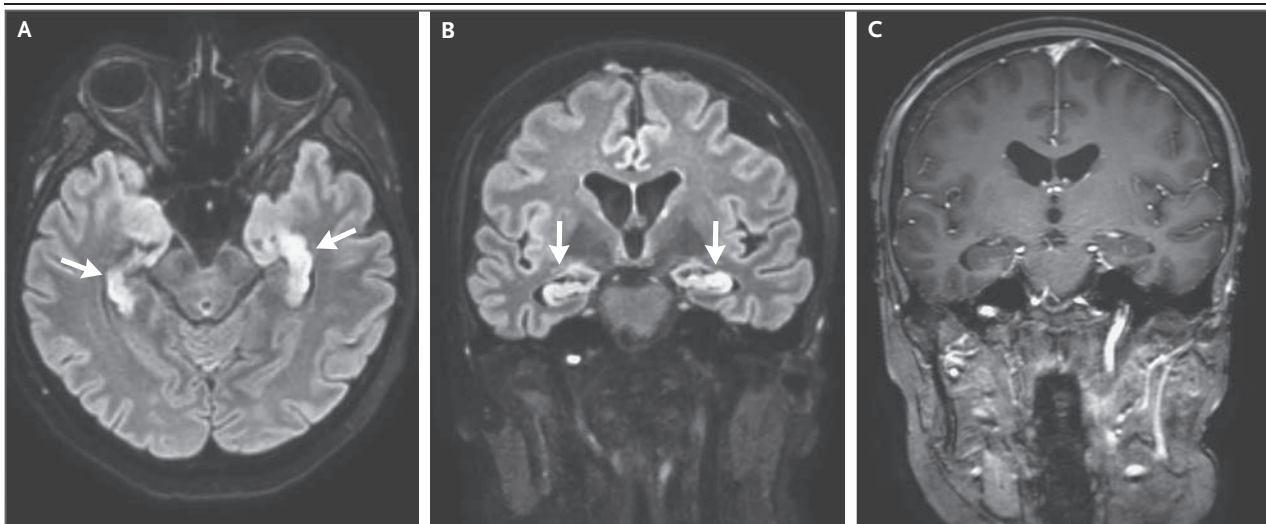


Figure 2. Cranial MRI 6 Weeks after CAR-T Infusion.

Axial (Panel A) and coronal (Panel B) FLAIR images show a hyperintense signal (arrows) in both hippocampi, and a coronal gadolinium-enhanced T1-weighted image (Panel C) with subtle contrast enhancement is consistent with encephalitis.

24 hours after discontinuation of apixaban, showed an opening pressure of 10 cm of water. The CSF was clear and colorless, with normal glucose and protein levels and no white cells. PCR testing of CSF was positive for HHV-6 DNA (10,000 copies per milliliter; cycle threshold, 32), but not for other pathogens. In the peripheral blood, HHV-6 DNA results were negative and the CMV load was 1200 IU per milliliter. Blood and CSF cultures remained negative. Tests for the presence in blood and CSF of autoantibodies against leucine-rich glioma-inactivated 1, contactin-associated protein–like 2, N-methyl-D-aspartate receptor, 8 dipeptidyl-peptidase–like protein 6, Hu, Yo, Ri, Tr, amphiphysin, anti-collapse response mediator protein 5, Ma1, and Ma2 were negative.

The MRI findings are consistent with limbic encephalitis, and the CSF studies show HHV-6 DNA and no evidence of autoimmune encephalitis or leptomeningeal lymphoma; these results support the diagnosis of HHV-6 encephalitis. CMV viremia, without signs of CMV disease, can be a result of cellular immunosuppression and is associated with HHV-6 encephalitis. In an immunocompetent patient with suspected acute viral encephalitis, acyclovir is routinely started, since herpes simplex virus is the most common causative

agent. However, in patients who are immunocompromised, empirical treatment with ganciclovir is recommended, with the addition of foscarnet if CMV encephalitis is suspected. In HHV-6 encephalitis, the role of dual therapy is unclear.

During the first week after antiviral treatment was initiated, the patient's clinical condition did not improve and focal epilepsy developed, which resolved with valproic acid. Foscarnet was discontinued after 1 week owing to severe electrolyte disturbances. The HHV-6 viral load in CSF cultures declined after 2 weeks (<1000 copies per milliliter; cycle threshold, 37), but the imaging abnormalities persisted. Treatment was discontinued after 3 weeks, when the presence of HHV-6 in CSF became undetectable. Because the patient had persisting memory impairment and disorientation 5 weeks after presentation, MRI was repeated and showed resolution of hyperintense lesions in the hippocampus with widening of the temporal horns as a result of hippocampal atrophy (Fig. 3). After 3 months of hospitalization, she was oriented but her memory impairment was still severe. She was transferred to a neuropsychiatric rehabilitation clinic.

Despite effective treatment resulting in an undetectable HHV-6 viral load in the CSF, the patient's

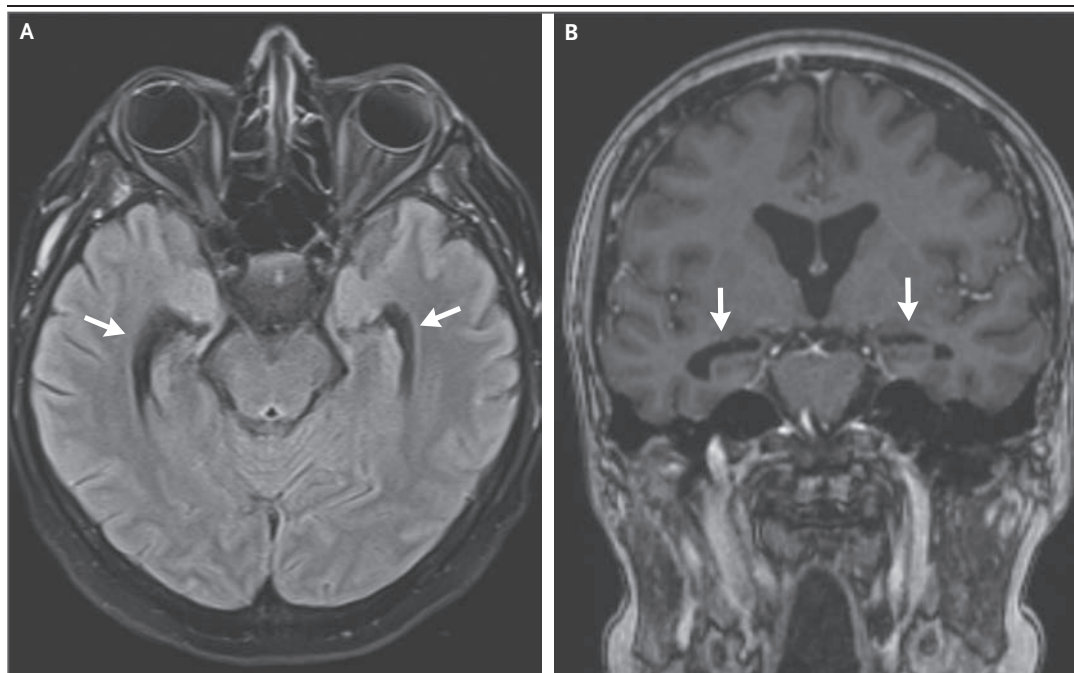


Figure 3. Cranial MRI 11 Weeks after CAR-T Infusion.

Axial (Panel A) FLAIR images and a coronal gadolinium-enhanced T1-weighted image (Panel B) show resolution of the hyperintense signal, with hippocampal atrophy resulting in enlargement of the temporal horns of the lateral ventricles (arrows).

neuropsychiatric symptoms improved only mildly even months after presentation, owing to hippocampal damage.

COMMENTARY

CD19-directed CAR-T therapy is increasingly used in the treatment of B-cell cancers. Its use results in 2-year progression-free survival in up to 40% of patients with relapsed or refractory diffuse large B-cell lymphoma¹; the prognosis for this patient population without CAR-T therapy is poor (2-year overall survival, 20%).² A major drawback of CAR-T therapy is the occurrence of severe adverse effects, as seen in this patient, in whom cytokine release syndrome followed by immune effector cell-associated neurotoxicity syndrome developed, which resolved during treatment with glucocorticoids; however, treatment led to a suspected glucocorticoid-induced psychosis. The diagnosis of glucocorticoid-induced psychosis was supported by the complete resolution of symptoms after the discontinuation of glucocorticoids. Then, weeks later, when the

patient presented with another episode of confusion, HHV-6 encephalitis was diagnosed.

The pathogenesis of immune effector cell-associated neurotoxicity syndrome is incompletely understood. It is hypothesized that systemic inflammation and circulating cytokines trigger the activation of endothelial cells and disruption of the blood-brain barrier, which in turn cause an inflammatory cascade involving CAR-T and other immune-competent cells within the CNS, subsequently affecting cortical and subcortical functions.³ Cytokine release syndrome is considered a cofactor in immune effector cell-associated neurotoxicity syndrome because, in the vast majority of patients, cytokine release syndrome occurs shortly before the development of immune effector cell-associated neurotoxicity syndrome.³ Both the incidence and severity of immune effector cell-associated neurotoxicity syndrome are higher with greater tumor volume, higher levels of LDH and inflammatory markers, lower platelet count, preceding cytokine release syndrome, and preexisting neurologic conditions.⁴ These risks also vary with the type of

CAR-T product and its costimulatory domain, with the highest incidence of severe neurotoxicity among patients who receive axicabtagene ciloleucel.⁵

When immune effector cell–associated neurotoxicity syndrome is suspected, immediate treatment with dexamethasone or methylprednisolone (at a dose of 1 to 2 g per day in severe disease) is recommended; tocilizumab is advised only if concurrent cytokine release syndrome is present.⁶ A cranial MRI to rule out cerebral edema or other causes of encephalopathy is indicated if immune effector cell–associated neurotoxicity syndrome does not resolve rapidly or if symptoms progress despite intensified glucocorticoid therapy. Most often, no abnormalities are seen on MRI in immune effector cell–associated neurotoxicity syndrome, although several nonspecific MRI patterns have been described.⁷ CSF analysis or an electroencephalogram (EEG) should be considered if symptoms do not resolve rapidly. CSF analysis may show low-grade pleiocytosis and increased protein. EEG monitoring commonly shows diffuse slowing as a nonspecific indicator of encephalopathy; epileptiform discharges and nonconvulsive status epilepticus have also infrequently been described.^{6,7} Most cases of immune effector cell–associated neurotoxicity syndrome are reversible, although rare fatal cases of cerebral edema and brain herniation despite treatment have been reported.⁵

HHV-6 belongs to the family of herpes viruses and affects virtually all young children as the cause of exanthema subitum. HHV-6 results in a chronic latent infection that can reactivate in the absence of immune surveillance, especially after allogeneic hematopoietic stem-cell transplantation or solid-organ transplantation.⁸ HHV-6 DNA in peripheral blood can reflect reactivation or a rare condition of inherited chromosomally integrated HHV-6. In patients who undergo hematopoietic stem-cell transplantation, reactivation of HHV-6 in the peripheral blood is usually asymptomatic; encephalitis develops in only 1% of patients with reactivation. HHV-6 DNA in CSF that coincides with typical symptoms and the exclusion of other causes is considered diagnostic of HHV-6 encephalitis.⁹

To our knowledge, eight cases of HHV-6 en-

cephalitis in CAR-T recipients have been reported to date.^{1,10-13} Patients with HHV-6 encephalitis typically present with acute memory loss, altered mental status, and seizures. Fever is uncommon. Mild CSF lymphocytic pleiocytosis, temporal abnormalities shown on EEG, and hyperintense lesions in the limbic system shown on MRI are typical findings.^{12,14} The preferred therapy for HHV-6 encephalitis is unknown. On the basis of inhibitory effects against HHV-6 shown in vitro and results reported in retrospective studies, ganciclovir or foscarnet is considered a first-line option. Double therapy can be considered in severely ill patients, although evidence for improved outcome with double therapy as compared with single therapy is lacking.^{14,15} Delayed initiation of antiviral treatment allows HHV-6 replication in the brain and results in rapidly progressive disease with convulsions and coma. Use of glucocorticoids can accelerate viral replication and should be discontinued.⁸ The recommended duration of therapy is 3 weeks, but therapy can be extended to 6 weeks if HHV-6 DNA remains detectable in CSF. Prophylactic antiviral therapy has been shown to be ineffective.¹⁴ In a series of recipients of hematopoietic stem-cell transplant who had HHV-6 encephalitis that was treated with antiviral therapy, the overall survival was approximately 60% after 100 days, with severe irreversible neurologic sequelae in 57% of survivors.¹⁵

Identifying the cause of confusion in patients treated with CAR-T therapy is challenging but crucial, since different diagnoses require distinct treatment strategies. The present case highlights the need for primary consideration of CAR-T–related neurologic toxicity and consideration of other diagnoses when the nature and timing of symptoms or responses to therapy are inconsistent with immune effector cell–associated neurotoxicity syndrome.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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